

circuitry. Photosensor noise mainly comprises Johnson noise, shot noise, generation-recombination noise and flicker noise.

Johnson noise, also known as Nyquist noise or thermal noise, is caused by thermal motion of charged particles in a resistive element. The RMS value of noise voltage depends on the resistance value, temperature and system bandwidth.

Shot noise in a photosensor is caused by the discrete nature of photoelectrons generated. It is related to the statistical fluctuation of both the dark current and the photo current. It depends on the average current through the photosensor and system bandwidth. It is the dominant source of noise in the case of photodiodes operating in photoconductive mode.

Generation-recombination noise is caused by fluctuation in current generation and recombination rates in a photosensor. This type of noise is predominant in photoconductive sensors operating at infrared wavelengths.

Flicker noise, or $1/f$ noise, occurs in all conductors where the conducting medium is not a metal and all semiconductor devices that require bias current for their operation. Its amplitude is inversely proportional to the frequency. Flicker noise is usually predominant at frequencies below 100Hz.

Photoconductors

Photoconductors—also referred to as photoresistors, light-dependent resistors (LDRs) and photocells—are semiconductor photosensors whose resistance decreases with increase in incident light intensity. These are bulk semiconductor devices with no p-n junction. The resistance change in a photoconductor is of the order of 6 decades, ranging from few tens of mega-ohms under dark conditions to few tens or hundreds of ohms under bright light conditions. Other features include wide dynamic response, spectral coverage from ultraviolet to far infrared, and low cost. However, photoconductors have response time of the order of hundreds of milliseconds.

Commonly used materials for photoconductors are cadmium-sulphide (CdS), lead-sulphide (PbS), lead-



Fig. 2: Some representative packages of photoconductors

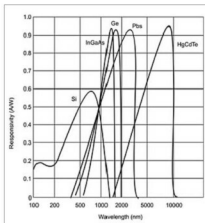


Fig. 3: Spectral characteristics of photodiodes

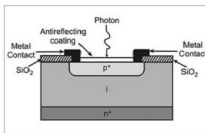


Fig. 4: PIN photodiode

selenide (PbSe), mercury-cadmium-telluride (HgCdTe) and germanium-copper (Ge:Cu). Inexpensive CdS photoconductors are used in many consumer items like camera light meters, clock radios, security alarms, street lights and so on. On the other hand, Ge:Cu cells are used for infrared astronomy and infrared spectroscopy. Fig. 2 shows some representative photoconductors.

Photoconductors are further classified as intrinsic or extrinsic depending upon whether an external dopant has been added or not to the semiconductor material. Intrinsic photoconductors operate at shorter wavelengths as electrons have to jump from valence to conduction band. Extrinsic photoconductors have a spectral response covering longer wavelengths.

A simple application circuit using a photoconductor would be a potential

divider arrangement comprising a fixed resistor and photoconductor. A fixed DC voltage is applied to the potential divider and the voltage appearing across either the fixed resistor or the photoconductor is taken as a measure of light intensity. A better option would be to use the photoconductor in conjunction with an operational amplifier wired either as a trans-impedance amplifier or voltage-mode amplifier. This gives better responsivity and higher output voltage.

Photodiodes

Photodiodes are junction-type semiconductor light sensors that generate current or voltage when p-n junction in the semiconductor is illuminated by light of sufficient energy. Spectral response of the photodiode is a function of the band-gap energy of the material used in its construction. Photodiodes are mostly constructed using silicon, germanium, indium-gallium-arsenide (InGaAs), lead-sulphide (PbS) and mercury-cadmium-telluride (HgCdTe). Fig. 3 shows spectral characteristics of these photodiodes.

Types of photodiodes. Depending upon construction, there are several types of photodiodes. These include p-n photodiodes, PIN photodiodes, Schottky-type photodiodes and avalanche photodiodes.

p-n photodiodes comprise a p-n junction. When light with sufficient energy strikes the photodiode, its electrons are pulled up into conduction band, leaving behind holes in valence band. These electron-hole pairs occur throughout p-layer, depletion layer and n-layer materials. When the photodiode is reverse biased, photo-induced electrons move down the potential hill from p-side to n-side. Similarly, photo-induced holes add to the current flow by moving across the junction to p-side. Shorter wavelengths are absorbed at the surface, while longer wavelengths penetrate deep into the diode. p-n photodiodes are used for precision photometry applications like medical instrumentation, analytical instruments, semiconductor tools and industrial measurement systems.